

Anomalous Magnetic Moments in FFGFT Theory

Geometric Derivation from Time-Mass Duality

Purely Geometric Formulas and Precise Ratio Predictions

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Contents

Abstract

In the present work, the fundamental architecture of spacetime is reinterpreted within the framework of **Fundamental Fractal Geometric Field Theory (FFGFT)** – internally referred to as the T0 model (B18). The central paradigm consists in the transition from a point-like to a purely geometric description of the vacuum as a four-dimensional **Gyral Torus**.

Geometric Structure: The theory is based on the fractal-geometric foundation with the parameter $\xi \approx (4/3) \times 10^{-4}$ and the densest local sphere packing by regular **Tetrahedra**. This tetrahedral basis forms the stable foundation for the low generations (electron, muon, proton/neutron) as well as the local 3D crystal structure of the torus. Building upon this, the ideal sub-Planck factor

$$f = 7500,$$

emerges through fractal branching and pentagonal symmetry breaking, representing an exactly 7500-fold reduction compared to the conventional Planck scale (t_0) and following directly from the geometric winding density $30000/4$.

g-2 Anomaly: A core element of the work is the transparent geometric derivation of the anomalous magnetic moments of leptons. While the Standard Model relies on numerous perturbative terms, in FFGFT the electron anomaly follows directly from the base winding (tetrahedral projection). The muon and tau anomalies arise from fractal branchings with Hausdorff dimensions $p \approx 5/3$ and $4/3$, respectively. With the ideal value $f = 7500$ and the geometric projection factor $k_{\text{geom}} = (2/\sqrt{\varphi})\sqrt{2} = 2.224$, the purely geometric predictions achieve an accuracy of about 2 %. Through the **fractal correction** $K_{\text{frak}}^{3/2}$, this systematic deviation is explained purely geometrically: the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2692$ agrees with the experimentally reconstructed value $k_{\text{rec}} = 2.2696$ to 0.01 %. The corrected absolute values achieve 0.01% (electron) and 0.15% (muon) accuracy. The most precise, k_{geom} -independent prediction for the tau anomaly is

$$a_\tau \approx 1.282 \times 10^{-3},$$

which follows exclusively from the exact ratio $f^{1/3} - 1$.

Geometric Proportionality: All physical base quantities (constants, masses, couplings) stand in fixed geometric ratios, drastically reducing the number of free parameters compared to the Standard Model. The T0 theory thus offers a complete geometric description and provides concrete, experimentally testable predictions – particularly for the tau anomaly as a decisive test at Belle II.

Note on Older Documents

Previous versions of the g-2 analysis (018_T0_Anomale-g2-9/10) used semi-empirical factors or the ideal k_{geom} without fractal correction. The present Rev. 11 shows that the 2 % deviation of absolute values is **fully explained by the fractal correction** $K_{\text{frak}}^{3/2}$ – the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$ agrees with the experimentally reconstructed value to 0.01 %.

Keywords: Anomalous magnetic moment, g-2, T0 theory, Time-Mass Duality, Torsion lattice, Ratio predictions, Koide formula

.1 Introduction: Geometric vs. Semi-Empirical Approaches

The Philosophy of T0 Theory

The T0 theory is based on the principle that **all** physical constants should follow from the geometric structure of a 4-dimensional torsion lattice. For anomalous magnetic moments this means:

- **NO** hidden fit parameters
- **ONLY** geometric factors: φ, ξ, f
- Honesty about precision limits
- Consistency with other predictions

Consistency with Mass Predictions

The T0 theory predicts lepton masses with 0.4–1.2% deviation:

Lepton	T0 [MeV]	Exp [MeV]	Deviation
Electron	0.505	0.511	1.18%
Muon	105.0	105.7	0.66%
Tau	1783	1777	0.37%

Table 1: Lepton masses in T0

Expectation: g-2 should have similar precision (1%).

With the fractal correction $K_{\text{frak}}^{3/2}$, the g-2 predictions achieve 0.01–0.15% accuracy, comparable to the mass precision.

.2 Physical Fundamentals

What is the Anomalous Magnetic Moment?

The magnetic moment of a charged spin-1/2 particle is:

$$\mu = g \cdot \frac{e}{2m} \cdot \frac{\hbar}{2} \quad (1)$$

where g is the gyromagnetic factor (g-factor).

Dirac Prediction: For a point-like particle: $g = 2$

Quantum Effects: Vacuum polarization, vertex corrections $\Rightarrow g \neq 2$

Anomaly: $a = (g - 2)/2$

QED Expectation: $a \approx \alpha/(2\pi) + \mathcal{O}(\alpha^2) \approx 0.00116$

T0 Interpretation: Windings in the Torsion Lattice

In T0 theory, leptons are **winding structures** in the 4D torsion lattice:

- **Electron:** Simple winding (1st generation)

- **Muon:** Winding with fractal branching (2nd generation)
- **Tau:** More complex fractal structure (3rd generation)
The anomalous moment arises from:
 1. The **rotation** of the winding (spin)
 2. The **charge distribution** on the winding
 3. The **projection** 4D → 3D
⇒ **No** point-like charge ⇒ $a \neq 0$

.3 Geometric Formulas

Fundamental Parameters

The T0 theory uses exclusively three geometric fundamental constants:

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.618 \dots \quad (\text{Golden Ratio}) \quad (2)$$

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (\text{Torsion constant}) \quad (3)$$

$$f = 7500 \quad (\text{Sub-Planck factor}) \quad (4)$$

The Real Sub-Planck Factor: $f = 7500$

Now we put everything together: The ideal crystal remains intact, the symmetry breaking only affects the projection factors:

$$\boxed{f = 7500} \quad (5)$$

This is the **most fundamental number of the T0 theory**. It appears in almost all formulas and describes:

- The number of Sub-Planck cells per Planck length
- The density of the torsion lattice
- The fundamental frequency of all geometric resonances

The Symmetry Breaking: The Role of the Golden Ratio

A perfect, ideal crystal would be completely symmetric. Yet our world shows symmetry breaking on all levels:

- Matter dominates over antimatter
- The weak interaction violates parity symmetry
- The neutron is heavier than the proton
- The three lepton generations have different masses

In the T0 theory, all these symmetry breakings have a single, geometric origin: the pentagonal symmetry of the crystal, embodied by the **golden ratio** φ . The golden ratio $\varphi = (1 + \sqrt{5})/2 = 1.618033989 \dots$ is the irrational number describing pentagonal symmetry. In a perfect pentagon, φ appears everywhere: The ratio of diagonal to side is exactly φ . Why pentagonal symmetry specifically? For deep mathematical reasons, pentagonal symmetry is the first one that **cannot tile the plane periodically**. This leads to *quasicrystals* – structures that are ordered but not periodic. Exactly such a quasicrystalline structure is postulated by

T0 theory for the Sub-Planck scale. The symmetry breaking is quantified in the theory not by a direct subtraction of 5φ from the ideal anchor number 7500. Instead, it manifests in the **fractal correction** $K_{\text{frak}}^{3/2}$, which corrects the ideal geometric projection factor k_{geom} to the effective value k_{eff} :

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \approx 2.22357, \quad (6)$$

which projects the 4D torsion onto the 3D world. The **fractal correction** yields the effective projection factor:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{0.98007} = 2.2692 \quad (7)$$

The exponent $3/2$ corresponds to the half spatial dimension ($D/2 = 3/2$) and describes how the fractal structure $K_{\text{frak}} = 1 - 100\xi = 0.98667$ acts across three spatial dimensions on the projection.

Crucially: The experimentally reconstructed value $k_{\text{geom}}^{\text{rec}} = 2.26955$ agrees with $k_{\text{eff}} = 2.2692$ to **0.014%** – the 2% deviation of the ideal formula is not a limitation but is fully explained by the fractal correction.

From the ideal 7500 remained the ideal 7500. This number became the new fundamental constant of the universe. It determined how densely the lattice was packed, how quickly torsion could propagate, which resonances were possible. Everything we observe today – every particle mass, every force strength, every cosmological constant – is a consequence of this single geometric story: From perfect crystal to pentagonally broken reality, with the breaking quantitatively captured by $K_{\text{frak}}^{3/2}$.

Electron: Base Winding

Formula:

$$a_e = \frac{S_3/f}{k_{\text{geom}}} \quad (8)$$

where:

- $S_3 = 2\pi^2 = 19.739$: 3D surface of the 4D winding
- $f = 7500$: Sub-Planck scaling
- k_{geom} : Geometric projection factor

Geometric Projection Factor:

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \quad (9)$$

Explanation of Factors:

- $2/\sqrt{\varphi} = 1.572$: Pentagonal projection (from ξ -structure)
- $\sqrt{2} = 1.414$: Diagonal projection 4D $\rightarrow$ 3D
- $k_{\text{geom}} = 2.224$: Completely geometric!

Numerical Calculation:

$$k_{\text{geom}} = \frac{2}{\sqrt{1.618}} \times \sqrt{2} = 2.224 \quad (10)$$

$$a_e = \frac{19.739/7500}{2.224} \quad (11)$$

$$a_e = 1.184 \times 10^{-3} \quad (12)$$

Comparison (ideal zeroth order):

- T0 (ideal): $a_e = 1.184 \times 10^{-3}$
- Experiment: $a_e = 1.160 \times 10^{-3}$
- Deviation: **2.03%**

With fractal correction $K_{\text{frak}}^{3/2}$:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{0.98007} = 2.2692 \quad (13)$$

$$a_e^{(\text{corr})} = \frac{S_3/f}{k_{\text{eff}}} = \frac{19.739/7500}{2.2692} = 1.1598 \times 10^{-3} \quad (14)$$

- T0 (corrected): $a_e = 1.1598 \times 10^{-3}$
- Experiment: $a_e = 1.1597 \times 10^{-3}$
- Deviation: **0.014%**

Muon: Fractal Additional Winding

Formula:

$$a_\mu = a_e + \Delta a_{\text{fractal}} \quad (15)$$

with

$$\Delta a_{\text{fractal}} = \frac{4\pi}{f^{p_\mu}} \quad (16)$$

where:

- $p_\mu = 5/3$: Fractal Hausdorff dimension
- 4π : Complete torsion revolution

Meaning of $p_\mu = 5/3$:

This is the well-known Hausdorff dimension of:

- Brownian motion in 2D
- Self-avoiding random walk
- Koch curve (fractal)

⇒ Physically plausible for “partially branched winding”!

Numerical Calculation:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{5/3}} = 4.373 \times 10^{-6} \quad (17)$$

$$a_\mu = 1.184 \times 10^{-3} + 4.373 \times 10^{-6} \quad (18)$$

$$a_\mu = 1.188 \times 10^{-3} \quad (19)$$

Comparison (ideal zeroth order):

- T0 (ideal): $a_\mu = 1.188 \times 10^{-3}$
- Experiment: $a_\mu = 1.166 \times 10^{-3}$
- Deviation: **1.89%**

With fractal correction:

$$a_\mu^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1598 \times 10^{-3} + 4.373 \times 10^{-6} \quad (20)$$

$$a_\mu^{(\text{corr})} = 1.1642 \times 10^{-3} \quad (21)$$

- T0 (corrected): $a_\mu = 1.1642 \times 10^{-3}$
- Experiment: $a_\mu = 1.1659 \times 10^{-3}$
- Deviation: **0.15%**

Tau: More Complex Fractal Structure

Formula:

$$a_\tau = a_e + \frac{4\pi}{f p_\tau} \quad (22)$$

where:

- $p_\tau = 4/3$: Stronger fractal branching

Meaning of $p_\tau = 4/3$:

This is the box-counting dimension of many fractals (e.g., Koch curve, Mandelbrot set).

Numerical Calculation:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{4/3}} = 8.560 \times 10^{-5} \quad (23)$$

$$a_\tau = 1.184 \times 10^{-3} + 8.560 \times 10^{-5} \quad (24)$$

$$a_\tau = 1.269 \times 10^{-3} \quad (25)$$

Status: This is a **prediction** – tau-g-2 has not been measured yet!

Corrected prediction:

$$a_\tau^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1598 \times 10^{-3} + 8.560 \times 10^{-5} \quad (26)$$

$$a_\tau^{(\text{corr})} = 1.2454 \times 10^{-3} \quad (27)$$

.4 Two Classes of Predictions: Absolute Values vs. Ratios

Why 2% Deviation for Absolute Values?

Where do the 2% in the ideal formula come from?

The 2% deviation of the ideal formula (with $k_{\text{geom}} = 2.224$) is **no longer an open question** but is fully explained by the fractal correction:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.224}{0.98007} = 2.2692 \quad (28)$$

Physical meaning:

- $K_{\text{frak}} = 1 - 100\xi = 0.98667$ is the universal fractal correction of T0 theory
- The exponent $3/2 = D/2$ (half spatial dimension) describes the effect of fractal structure on the 4D→3D projection
- This correction is **not a fit**: K_{frak} follows directly from ξ and already appears in the mass formulas
- The experimentally reconstructed value $k_{\text{rec}} = 2.2696$ confirms the correction to 0.014%

Consistency: The same fractal correction K_{frak} appears in:

1. Particle masses: $m_i = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$
2. Gravitational constant: $G = f(\xi, K_{\text{frak}})$
3. g-2 anomalies: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$

Thus T0 theory is fully consistent: **one** correction parameter K_{frak} with **different** exponents depending on the physical context.

Ratios are Mathematically Exact

In contrast to absolute values, **ratios of differences** are structurally exact:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} = f^{1/3} - 1 \quad (29)$$

Why is this exact?

- The common factor 4π cancels out
- The projection factor k_{geom} cancels out
- Only the fractal exponents (5/3 and 4/3) determine the ratio
- The result depends **only** on f : $f^{1/3} - 1 = 18.57$

Important

Fundamental Distinction **Absolute values**:

- Depend on k_{geom} , f_i and SI conversion
- 2% deviation due to higher-order quantum effects
- Consistent with all T0 predictions

Ratios:

- Depend **only** on f
- k_{geom} and SI factors cancel out
- Mathematically exact from fractal exponents
- Difference $< 10^{-13}$ (numerical precision)

⇒ The ratio prediction is **not an approximation**, but an **exact geometric relation**!

Analogy to the Koide Formula

The Koide formula for lepton masses:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.666661 \approx \frac{2}{3} \quad (30)$$

confirmed to 0.001%, shows: **ratios** are more precise than absolute values! In T0 it emerges from the exponent structure $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ with step $\Delta p_{\mu \rightarrow \tau} = 1/3 = 1/D$ (the step $e \rightarrow \mu$ is $1/2$ — the special status of the electron).

The same $1/3$ -step appears in the $g-2$ exponents: $q_\mu = 5/3$, $q_\tau = 4/3$, $\Delta q = 1/3$. This is no coincidence — it is the same geometric origin.

The bridge from masses to $g-2$: From the mass formulas follows $m_\tau/m_\mu = (125/144) \cdot f^{1/3}$, from the $g-2$ formulas $\Delta a(\tau-e)/\Delta a(\mu-e) = f^{1/3}$. Combining yields:

$$\boxed{\frac{\Delta a(\tau-e)}{\Delta a(\mu-e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu}} \quad (31)$$

The $g-2$ ratio is determined by the mass ratio. No α , no k_{eff} , no K_{frak} — everything cancels out.

For $g-2$ in T0:

- **Absolute values:** 2% deviation
- **Ratio** $\Delta a(\tau - \mu)/\Delta a(\mu - e)$: Exactly $= f^{1/3} - 1$

- **Bridge to masses:** $\Delta a(\tau-e)/\Delta a(\mu-e) = (144/125) \cdot m_\tau/m_\mu$

This is **not a weakness**, but shows the **geometric structure** of the theory!

.5 Precise Ratio Predictions

Analogy to the Koide Formula

The Koide formula for lepton masses:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \pm 0.0004\% \quad (32)$$

shows: **Ratios** are more precise than absolute values!

Question: Does this also hold for g-2?

The Ratio of Differences

Define the differences:

$$\Delta a(\mu - e) = a_\mu - a_e = \frac{4\pi}{f^{5/3}} \quad (33)$$

$$\Delta a(\tau - \mu) = a_\tau - a_\mu = \frac{4\pi}{f^{4/3}} - \frac{4\pi}{f^{5/3}} \quad (34)$$

Ratio:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} \quad (35)$$

$$= \frac{f^{5/3}}{f^{4/3}} - 1 \quad (36)$$

$$= f^{5/3-4/3} - 1 \quad (37)$$

$$= f^{1/3} - 1 \quad (38)$$

Important

Core Prediction

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = f^{1/3} - 1 = 18.57 \quad (39)$$

This relation is:

- **Parameter-free** (only f !)
- **Independent** of k_{geom}
- **Exact** (difference $< 10^{-13}$)
- **Testable** at Belle II

Numerical Verification

With $f = 7500$:

$$f^{1/3} = 7500^{1/3} = 19.57 \quad (40)$$

$$f^{1/3} - 1 = 18.57 \quad (41)$$

From T0 values:

$$\Delta a(\mu - e) = 4.373 \times 10^{-6} \quad (42)$$

$$\Delta a(\tau - \mu) = 8.123 \times 10^{-5} \quad (43)$$

$$\text{Ratio} = \frac{8.123 \times 10^{-5}}{4.373 \times 10^{-6}} = 18.57 \quad (44)$$

Agreement: Perfect! ✓✓✓

Testable Prediction for Tau

The prediction follows in one step from the bridge formula and three experimental numbers:

$$\Delta a(\mu - e)^{\text{exp}} = a_\mu - a_e = 6.269 \times 10^{-6} \quad (45)$$

$$\frac{m_\tau}{m_\mu} = 16.817 \quad (46)$$

Prediction:

$$\Delta a(\tau - e) = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} \cdot \Delta a(\mu - e)^{\text{exp}} \quad (47)$$

$$= 1.152 \times 16.817 \times 6.269 \times 10^{-6} = 1.214 \times 10^{-4} \quad (48)$$

$$a_\tau^{\text{predicted}} = a_e + \Delta a(\tau - e) \quad (49)$$

$$= 1.160 \times 10^{-3} + 1.214 \times 10^{-4} \quad (50)$$

$$= \boxed{1.282 \times 10^{-3}} \quad (51)$$

That is all. No α , no k_{eff} , no K_{frak} . Only the bridge formula applied to measured values.

Consistency check: The SM calculates $a_\tau^{\text{SM}} = 1.177 \times 10^{-3}$, corresponding to a ratio $\Delta a(\tau - e)/\Delta a(\mu - e) = 2.80$. T0 predicts 19.4 — a factor of 7, clearly testable at Belle II.

.6 The Fractal Correction $K_{\text{frak}}^{3/2}$: Resolving the 2% Deviation

Central Insight

The 2% deviation of the ideal g-2 formula is **identical** to the fractal correction $K_{\text{frak}}^{3/2}$, already known from the mass formulas:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{(2/\sqrt{\varphi})\sqrt{2}}{(1 - 100\xi)^{3/2}} = \frac{2.224}{0.98007} = 2.2692 \quad (52)$$

The experimentally reconstructed value $k_{\text{rec}} = 2.269\%$ confirms this to **0.014%**.

Why the Exponent $3/2$?

The exponent $3/2 = D/2$ has a clear geometric meaning:

- The fractal correction $K_{\text{frak}} = 1 - 100\xi$ describes the deviation of the real lattice from the ideal crystal structure
- In the 4D→3D projection (k_{geom}), this correction acts across $D = 3$ spatial dimensions
- Per dimension: $K_{\text{frak}}^{1/2}$ (square root of the 1D correction)
- Total: $K_{\text{frak}}^{D/2} = K_{\text{frak}}^{3/2}$

Consistency with Other T0 Corrections

Physical Quantity	K_{frak} Exponent	Interpretation
Particle masses	K_{frak}^1	Scalar correction (1D)
g-2 projection factor	$K_{\text{frak}}^{3/2}$	3D spatial projection ($D/2$)
Gravitational constant	K_{frak}^1	Scalar correction (1D)

Table 2: Universal fractal correction with context-dependent exponent

The different exponents are **not free parameters** but follow from the dimensionality of the respective physical process.

.7 The Bridge Formula: Three Numbers, One Prediction

The entire T0 complexity cancels out. The tau prediction needs **only**:

1. a_e and a_μ (experimentally measured)
2. $m_\tau/m_\mu = 16.817$ (experimental, high-precision)
3. The rational factor 144/125 from T0 geometry

The Calculation

Step 1: Measured difference

$$\Delta a(\mu-e) = a_\mu - a_e = 6.269 \times 10^{-6} \quad (53)$$

Step 2: Apply bridge formula (Eq. ??)

$$\frac{\Delta a(\tau-e)}{\Delta a(\mu-e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} = 1.152 \times 16.817 = 19.37 \quad (54)$$

Step 3: Result

$$\Delta a(\tau-e) = 19.37 \times 6.269 \times 10^{-6} = 1.214 \times 10^{-4} \quad (55)$$

$$a_\tau = a_e + \Delta a(\tau-e) = \boxed{1.281 \times 10^{-3}} \quad (56)$$

That is all. No α , no k_{eff} , no K_{frak} , no perturbation theory, no 12,672 Feynman diagrams.

Why So Simple?

T0 theory gives $\Delta a_i = 4\pi/f^{q_i}$, from which $\Delta a(\tau-e)/\Delta a(\mu-e) = f^{1/3}$ follows. From the masses follows $m_\tau/m_\mu = (125/144) \cdot f^{1/3}$. Combining:

$$\frac{\Delta a(\tau-e)}{\Delta a(\mu-e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} \quad (57)$$

The parameter f cancels out. The prediction is **independent** of the exact value of f , ξ , k_{eff} , or any other T0 parameter. It follows solely from:

1. The existence of a 1/3-step (Koide structure)
2. The fact that this step appears in both masses and g-2

Falsifiability

Even those skeptical of T0 theory as a whole can test this one relation:

- Belle II measures $a_\tau \approx 1.18 \times 10^{-3}$: Relation **falsified**.
- Belle II measures $a_\tau \approx 1.28 \times 10^{-3}$: Koide-g-2 connection **confirmed** — independent of the rest of the theory.

.8 Experimental Tests

Belle II (2027–2028)

Belle II expects sensitivity of $\sim 10^{-7}$ for a_τ .

Test 1: Absolute value

- T0 prediction (corrected): $a_\tau = 1.245 \times 10^{-3}$
- T0 from bridge formula: $a_\tau = 1.282 \times 10^{-3}$
- SM: $a_\tau = 1.177 \times 10^{-3}$
- Difference T0 vs. SM: 9%
- Difference: 1%

Test 2: Ratio

- T0 prediction: $\Delta a(\tau - \mu) / \Delta a(\mu - e) = 18.57$
- This is the **more precise** prediction!
- Independent of absolute calibration

Possible outcomes:

1. **Confirmation:** Ratio ≈ 18.6
⇒ Strong evidence for fractal structure hypothesis
2. **Deviation:** Ratio $\neq 18.6$
⇒ Different fractal dimensions or additional physics
3. **Null result:** $a_\tau < 10^{-8}$
⇒ T0 contributions suppressed or theory needs revision

Fermilab/J-PARC

Further precision improvements for a_μ :

- Reduction of experimental uncertainties
- Clearer determination of SM discrepancy
- Refinement of $\Delta a(\mu - e)$ measurement

.9 Comparison with Other Approaches

T0 Philosophy: We choose **explainability** over precision!

Approach	Precision	Parameters	Explainable
QED (SM)	Perfect	Many	Yes
T0 (ideal, k_{geom})	2%	0	Completely
T0 (with $K_{\text{frak}}^{3/2}$)	0.01–0.15%	0	Completely

Table 3: Comparison of different approaches

.10 Proof: The Experimental Reconstruction IS the Fractal Correction

The Central Insight

The ratio $\Delta a(\tau - \mu)/\Delta a(\mu - e) = f^{1/3} - 1$ is **mathematically exact** because the correction factor k_{geom} cancels out completely.

In Rev. 10, the experimentally reconstructed value $k_{\text{rec}} = 2.269$ was considered "semi-empirical." Now it turns out: this value is **exactly the fractal correction**:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.224}{0.98007} = 2.2692 \quad (58)$$

Comparison: Reconstructed vs. Fractal Corrected

$$k_{\text{geom}}^{(\text{reconstructed})} = \frac{S_3/f}{a_e^{(\text{exp})}} = \frac{2\pi^2/7500}{1.160 \times 10^{-3}} = 2.2696 \quad (59)$$

Comparison:

- Geometrically ideal: $k_{\text{geom}} = (2/\sqrt{\varphi}) \times \sqrt{2} = 2.224$
- Fractal corrected: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2692$
- Reconstructed from experiment: $k_{\text{geom}}^{(\text{rec})} = 2.2696$
- **Difference k_{eff} vs. k_{rec} : 0.014%**

This proves: The "semi-empirical" reconstruction was all along the **purely geometric** fractal correction $K_{\text{frak}}^{3/2}$.

Results with Fractal Correction

Lepton	Ideal ($k = 2.224$)	Corr. ($k_{\text{eff}} = 2.269$)	Experiment	Dev.
Electron	1.184×10^{-3}	1.1598×10^{-3}	1.1597×10^{-3}	0.01%
Muon	1.188×10^{-3}	1.1642×10^{-3}	1.1659×10^{-3}	0.15%
Tau	1.269×10^{-3}	1.2454×10^{-3}	(not measured)	Prediction

Table 4: Absolute values with geometric vs. fractal corrected k_{eff}

Important

Crucial Point The 2% deviation of the ideal formula is **fully** explained by $K_{\text{frak}}^{3/2}$:

- Electron: 2.03% $\rightarrow$ 0.014% (factor 145 improvement)
 - Muon: 1.89% $\rightarrow$ 0.15% (factor 13 improvement)
 - The fractal correction is **not a new parameter** but follows from $\xi = 4/30000$
- Thus the T0 g-2 calculation is **fully parameter-free** with a precision of 0.01–0.15%.

Alternative: Directly from Ratio Relation

Even more precise is the calculation directly from the exact ratio:

$$\Delta a(\mu - e)^{(\text{exp})} = a_{\mu}^{(\text{exp})} - a_e^{(\text{exp})} = 6.269 \times 10^{-6} \quad (60)$$

$$\Delta a(\tau - \mu) = \Delta a(\mu - e)^{(\text{exp})} \times (f^{1/3} - 1) \quad (61)$$

$$= 6.269 \times 10^{-6} \times 18.57 = 1.164 \times 10^{-4} \quad (62)$$

$$a_{\tau}^{(\text{Ratio})} = a_{\mu}^{(\text{exp})} + \Delta a(\tau - \mu) \quad (63)$$

$$= 1.166 \times 10^{-3} + 1.164 \times 10^{-4} \quad (64)$$

$$= \boxed{1.282 \times 10^{-3}} \quad (65)$$

Note: This prediction is **independent** of k_{geom} and uses only the exact geometric ratio structure!

Two Complementary Tau Predictions

Method	a_{τ} Prediction	Dependent on
Purely geometric	1.269×10^{-3}	$k_{\text{geom}} = 2.224$ (geometric)
With rec. k_{geom}	1.246×10^{-3}	$k_{\text{geom}} = 2.269$ (from a_e)
From ratio/bridge	1.282×10^{-3}	Only f and exp. masses (exact)
Range	$1.25\text{--}1.28 \times 10^{-3}$	$\pm 1.5\%$

Table 5: Three T0 predictions for a_{τ}

What does this mean for Belle II?**If Belle II measures:**

1. $a_{\tau} \approx 1.28 \times 10^{-3}$:
 - ✓ Confirms the exact ratio relation $f^{1/3} - 1$
 - ✓ Shows that experimental a_{μ} and ratio structure are correct
 - $\rightarrow$ **Strongest confirmation of T0 geometry**
2. $a_{\tau} \approx 1.25 \times 10^{-3}$:
 - ✓ Confirms fractal correction $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$
 - ✓ Shows that the 2% deviation is fully explained by $K_{\text{frak}}^{3/2}$

- → **Strongest confirmation of fractal T0 geometry**
3. $a_\tau \approx 1.27\text{--}1.28 \times 10^{-3}$:
 - ✓ Confirms exact ratio relation $f^{1/3} - 1$
 - ? Fractal correction at tau with different exponent?
 4. a_τ **outside** 1.24–1.28:
 - × T0 structure needs revision

Key Statement

The 2% deviation of the purely geometric T0 predictions is **fully** explained by the fractal correction $K_{\text{frak}}^{3/2}$:

- Electron: $2.03\% \rightarrow 0.014\%$
- Muon: $1.89\% \rightarrow 0.15\%$

The effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2692$ is **not a new free parameter** but follows directly from $\xi = 4/30000$.

The most precise T0 prediction for tau uses the exact ratio relation:

$$a_\tau = 1.282 \times 10^{-3} \quad (66)$$

.11 Important Note: No α in the T0 g-2 Formulas

IMPORTANT: The T0 formulas for g-2 contain **no** α !

In natural units ($\hbar = c = \alpha = 1$):

$$a_\ell = f(\varphi, \xi, f, \text{generation quantum numbers})$$

The anomalous moment is a **purely geometric quantity**, following from the winding structure in the torsion lattice.

Ratios like $\Delta a(\tau - \mu)/\Delta a(\mu - e) = f^{1/3} - 1$ are **independent** of:

- α (fine-structure constant)
- SI conversion factors
- k_{geom} (projection factor)

They depend **ONLY** on the fractal structure!

T0 derives α : While the SM takes the fine-structure constant from experiment ($\alpha_{\text{exp}} = 1/137.036$), T0 computes it directly: $\alpha = \xi \cdot E_0^2$ with $E_0 = 7.398$ MeV gives $1/\alpha_{\text{T0}} = 137.035$ — only 0.0005% from experiment.

Fair Comparison: SM vs. T0

A common objection is that the SM computes a_e to 10^{-12} , while T0 achieves only 0.014%. This comparison is misleading:

- **SM circularity:** The SM extracts α from the a_e measurement and then computes a_e with this α . For electron and muon, this is a circular fit — not an independent prediction.
- **T0 derives α :** From ξ alone, without measurement. The 0.0005% deviation is a genuine, independent prediction.
- **Where the theories diverge:** Not at e or μ , but at the **tau**. T0: $a_\tau \approx 1.282 \times 10^{-3}$, SM: $a_\tau \approx 1.177 \times 10^{-3}$ — 9% difference, testable at Belle II.

Further Reading and Resources

T0 Theory and Python Scripts:

- Repository: github.com/jpascher/T0-Time-Mass-Duality
- Python scripts: github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/python/
- Time-Mass Duality documentation
- Fundamental Fractal Geometric Field Theory (FFGFT)

Experimental Results:

- Fermilab Muon g-2 (2025): muon-g-2.fnal.gov
- Theory Initiative White Paper
- Belle II: www.belle2.org

Related T0 Documents:

- Lepton masses: Systematic derivation from quantum numbers
- Koide formula in T0: Geometric interpretation
- Fractal spacetime: $D_f = 3 - \xi$